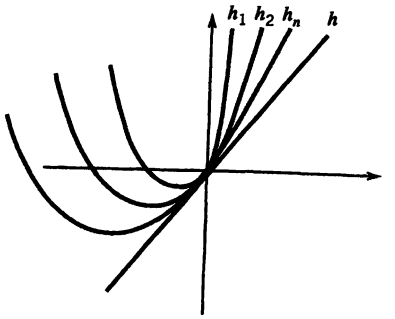


Figure 8.1.3 $h_n(x) = (x^2 + nx)/n$

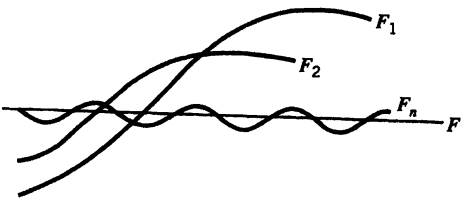


Figure 8.1.4 $F_n(x) = \sin(nx + n)/n$

(d) $\lim((1/n)\sin(nx + n)) = 0$ for $x \in \mathbb{R}$.

Let $F_n(x) := (1/n)\sin(nx + n)$ for $x \in \mathbb{R}$, $n \in \mathbb{N}$, and let $F(x) := 0$ for $x \in \mathbb{R}$. (See Figure 8.1.4.) Since $|\sin y| \leq 1$ for all $y \in \mathbb{R}$ we have

$$|F_n(x) - F(x)| = \left|\frac{1}{n}\sin(nx + n)\right| \leq \frac{1}{n} \tag{2}$$

for all $x \in \mathbb{R}$. Therefore it follows that $\lim(F_n(x)) = 0 = F(x)$ for all $x \in \mathbb{R}$. The reader should note that, given any $\varepsilon > 0$, if n is sufficiently large, then $|F_n(x) - F(x)| < \varepsilon$ for all values of x simultaneously! □

Partly to reinforce Definition 8.1.1 and partly to prepare the way for the important notion of uniform convergence, we reformulate Definition 8.1.1 as follows.

8.1.3 Lemma *A sequence (f_n) of functions on $A \subseteq \mathbb{R}$ to $\mathbb{R}$ converges to a function $f : A_0 \to \mathbb{R}$ on A_0 if and only if for each $\varepsilon > 0$ and each $x \in A_0$ there is a natural number $K(\varepsilon, x)$ such that if $n \geq K(\varepsilon, x)$, then*

$$|f_n(x) - f(x)| < \varepsilon. \tag{3}$$

We leave it to the reader to show that this is equivalent to Definition 8.1.1. We wish to emphasize that the value of $K(\varepsilon, x)$ will depend, in general, on *both* $\varepsilon > 0$ and $x \in A_0$. The reader should confirm the fact that in Examples 8.1.2(a–c), the value of $K(\varepsilon, x)$ required to obtain an inequality such as (3) does depend on both $\varepsilon > 0$ and $x \in A_0$. The intuitive reason for this is that the convergence of the sequence is "significantly faster" at some points than it is at others. However, in Example 8.1.2(d), as we have seen in inequality (2), if we choose n sufficiently large, we can make $|F_n(x) - F(x)| < \varepsilon$ for *all* values of $x \in \mathbb{R}$. It is precisely this rather subtle difference that distinguishes between the notion of the "pointwise convergence" of a sequence of functions (as defined in Definition 8.1.1) and the notion of "uniform convergence."

Uniform Convergence

8.1.4 Definition A sequence (f_n) of functions on $A \subseteq \mathbb{R}$ to $\mathbb{R}$ **converges uniformly on** $A_0 \subseteq A$ to a function $f : A_0 \to \mathbb{R}$ if for each $\varepsilon > 0$ there is a natural number $K(\varepsilon)$ (depending on ε but **not** on $x \in A_0$) such that if $n \geq K(\varepsilon)$, then

$$|f_n(x) - f(x)| < \varepsilon \qquad \text{for all} \quad x \in A_0. \tag{4}$$

In this case we say that the sequence (f_n) is **uniformly convergent on** A_0. Sometimes we write

$$f_n \rightrightarrows f \quad \text{on} \quad A_0, \qquad \text{or} \qquad f_n(x) \rightrightarrows f(x) \qquad \text{for} \quad x \in A_0.$$

It is an immediate consequence of the definitions that if the sequence (f_n) is uniformly convergent on A_0 to f, then this sequence also converges pointwise on A_0 to f in the sense of Definition 8.1.1. That the converse is not always true is seen by a careful examination of Examples 8.1.2(a–c); other examples will be given below.

It is sometimes useful to have the following necessary and sufficient condition for a sequence (f_n) *to fail* to converge uniformly on A_0 to f.

8.1.5 Lemma *A sequence (f_n) of functions on $A \subseteq \mathbb{R}$ to $\mathbb{R}$ does not converge uniformly on $A_0 \subseteq A$ to a function $f : A_0 \to \mathbb{R}$ if and only if for some $\varepsilon_0 > 0$ there is a subsequence (f_{n_k}) of (f_n) and a sequence (x_k) in A_0 such that*

$$|f_{n_k}(x_k) - f(x_k)| \geq \varepsilon_0 \qquad \text{for all} \quad k \in \mathbb{N}. \tag{5}$$

The proof of this result requires only that the reader negate Definition 8.1.4; we leave this to the reader as an important exercise. We now show how this result can be used.

8.1.6 Examples **(a)** Consider Example 8.1.2(a). If we let $n_k := k$ and $x_k := k$, then $f_{n_k}(x_k) = 1$ so that $|f_{n_k}(x_k) - f(x_k)| = |1 - 0| = 1$. Therefore the sequence (f_n) does not converge uniformly on $\mathbb{R}$ to f.

(b) Consider Example 8.1.2(b). If $n_k := k$ and $x_k := \left(\frac{1}{2}\right)^{1/k}$, then

$$|g_{n_k}(x_k) - g(x_k)| = \left|\tfrac{1}{2} - 0\right| = \tfrac{1}{2}.$$

Therefore the sequence (g_n) does not converge uniformly on $(-1, 1]$ to g.

(c) Consider Example 8.1.2(c). If $n_k := k$ and $x_k := -k$, then $h_{n_k}(x_k) = 0$ and $h(x_k) = -k$ so that $|h_{n_k}(x_k) - h(x_k)| = k$. Therefore the sequence (h_n) does not converge uniformly on $\mathbb{R}$ to h. □

The Uniform Norm

In discussing uniform convergence, it is often convenient to use the notion of the uniform norm on a set of bounded functions.

8.1.7 Definition If $A \subseteq \mathbb{R}$ and $\varphi : A \to \mathbb{R}$ is a function, we say that φ is **bounded on** A if the set $\varphi(A)$ is a bounded subset of $\mathbb{R}$. If φ is bounded we define the **uniform norm of** φ **on** A by

$$\|\varphi\|_A := \sup\{|\varphi(x)| : x \in A\}. \tag{6}$$

Note that it follows that if $\varepsilon > 0$, then

$$\|\varphi\|_A \leq \varepsilon \iff |\varphi(x)| \leq \varepsilon \qquad \text{for all} \quad x \in A. \tag{7}$$

8.1.8 Lemma *A sequence (f_n) of bounded functions on $A \subseteq \mathbb{R}$ converges uniformly on A to f if and only if $\|f_n - f\|_A \to 0$.*

Proof. ($\Rightarrow$) If (f_n) converges uniformly on A to f, then by Definition 8.1.4, given any $\varepsilon > 0$ there exists $K(\varepsilon)$ such that if $n \geq K(\varepsilon)$ and $x \in A$ then

$$|f_n(x) - f(x)| \leq \varepsilon.$$

From the definition of supremum, it follows that $||f_n - f||_A \leq \varepsilon$ whenever $n \geq K(\varepsilon)$. Since $\varepsilon > 0$ is arbitrary this implies that $||f_n - f||_A \to 0$.

($\Leftarrow$) If $||f_n - f||_A \to 0$, then given $\varepsilon > 0$ there is a natural number $H(\varepsilon)$ such that if $n \geq H(\varepsilon)$ then $||f_n - f||_A \leq \varepsilon$. It follows from (7) that $|f_n(x) - f(x)| \leq \varepsilon$ for all $n \geq H(\varepsilon)$ and $x \in A$. Therefore (f_n) converges uniformly on A to f. Q.E.D.

We now illustrate the use of Lemma 8.1.8 as a tool in examining a sequence of bounded functions for uniform convergence.

8.1.9 Examples **(a)** We cannot apply Lemma 8.1.8 to the sequence in Example 8.1.2(a) since the function $f_n(x) - f(x) = x/n$ is not bounded on $\mathbb{R}$.

For the sake of illustration, let $A := [0, 1]$. Although the sequence (x/n) did not converge uniformly on $\mathbb{R}$ to the zero function, we shall show that the convergence is uniform on A. To see this, we observe that

$$||f_n - f||_A = \sup\{|x/n - 0| : 0 \leq x \leq 1\} = \frac{1}{n}$$

so that $||f_n - f||_A \to 0$. Therefore (f_n) is uniformly convergent on A to f.

(b) Let $g_n(x) := x^n$ for $x \in A := [0, 1]$ and $n \in \mathbb{N}$, and let $g(x) := 0$ for $0 \leq x < 1$ and $g(1) := 1$. The functions $g_n(x) - g(x)$ are bounded on A and

$$||g_n - g||_A = \sup\begin{Bmatrix} x^n & \text{for} & 0 \leq x < 1 \\ 0 & \text{for} & x = 1 \end{Bmatrix} = 1$$

for any $n \in \mathbb{N}$. Since $||g_n - g||_A$ does *not* converge to 0, we infer that the sequence (g_n) does *not* converge uniformly on A to g.

(c) We cannot apply Lemma 8.1.8 to the sequence in Example 8.1.2(c) since the function $h_n(x) - h(x) = x^2/n$ is not bounded on $\mathbb{R}$.

Instead, let $A := [0, 8]$ and consider

$$||h_n - h||_A = \sup\{x^2/n : 0 \leq x \leq 8\} = 64/n.$$

Therefore, the sequence (h_n) converges uniformly on A to h.

(d) If we refer to Example 8.1.2(d), we see from (2) that $||F_n - F||_{\mathbb{R}} \leq 1/n$. Hence (F_n) converges uniformly on $\mathbb{R}$ to F.

(e) Let $G(x) := x^n(1 - x)$ for $x \in A := [0, 1]$. Then the sequence $(G_n(x))$ converges to $G(x) := 0$ for each $x \in A$. To calculate the uniform norm of $G_n - G = G_n$ on A, we find the derivative and solve

$$G_n'(x) = x^{n-1}(n - (n+1)x) = 0$$

to obtain the point $x_n := n/(n+1)$. This is an interior point of [0, 1], and it is easily verified by using the First Derivative Test 6.2.8 that G_n attains a maximum on [0, 1] at x_n. Therefore, we obtain

$$||G_n||_A = G_n(x_n) = (1 + 1/n)^{-n} \cdot \frac{1}{n+1},$$

which converges to $(1/e) \cdot 0 = 0$. Thus we see that convergence is uniform on A. □

By making use of the uniform norm, we can obtain a necessary and sufficient condition for uniform convergence that is often useful.

8.1.10 Cauchy Criterion for Uniform Convergence *Let (f_n) be a sequence of bounded functions on $A \subseteq \mathbb{R}$. Then this sequence converges uniformly on A to a bounded function f if and only if for each $\varepsilon > 0$ there is a number $H(\varepsilon)$ in $\mathbb{N}$ such that for all $m, n \geq H(\varepsilon)$, then $||f_m - f_n||_A \leq \varepsilon$.* □

Proof. ($\Rightarrow$) If $f_n \rightrightarrows f$ on A, then given $\varepsilon > 0$ there exists a natural number $K(\frac{1}{2}\varepsilon)$ such that if $n \geq K(\frac{1}{2}\varepsilon)$ then $||f_n - f||_A \leq \frac{1}{2}\varepsilon$. Hence, if both $m, n \geq K(\frac{1}{2}\varepsilon)$, then we conclude that

$$|f_m(x) - f_n(x)| \leq |f_m(x) - f(x)| + |f_n(x) - f(x)| \leq \tfrac{1}{2}\varepsilon + \tfrac{1}{2}\varepsilon = \varepsilon$$

for all $x \in A$. Therefore $||f_m - f_n||_A \leq \varepsilon$ for $m, n \geq K(\frac{1}{2}\varepsilon) =: H(\varepsilon)$.

($\Leftarrow$) Conversely, suppose that for $\varepsilon > 0$ there is $H(\varepsilon)$ such that if $m, n \geq H(\varepsilon)$, then $||f_m - f_n||_A \leq \varepsilon$. Therefore, for each $x \in A$ we have

$$|f_m(x) - f_n(x)| \leq ||f_m - f_n||_A \leq \varepsilon \qquad \text{for} \quad m, n \geq H(\varepsilon). \tag{8}$$

It follows that $(f_n(x))$ is a Cauchy sequence in $\mathbb{R}$; therefore, by Theorem 3.5.5, it is a convergent sequence. We define $f : A \to \mathbb{R}$ by

$$f(x) := \lim(f_n(x)) \qquad \text{for} \quad x \in A.$$

If we let $n \to \infty$ in (8), it follows from Theorem 3.2.6 that for each $x \in A$ we have

$$|f_m(x) - f(x)| \leq \varepsilon \qquad \text{for} \quad m \geq H(\varepsilon).$$

Therefore the sequence (f_n) converges uniformly on A to f. Q.E.D.

Exercises for Section 8.1

1. Show that $\lim(x/(x + n)) = 0$ for all $x \in \mathbb{R}, x \geq 0$.
2. Show that $\lim(nx/(1 + n^2x^2)) = 0$ for all $x \in \mathbb{R}$.
3. Evaluate $\lim(nx/(1 + nx))$ for $x \in \mathbb{R}$, $x \geq 0$.
4. Evaluate $\lim(x^n/(1 + x^n))$ for $x \in \mathbb{R}$, $x \geq 0$.
5. Evaluate $\lim((\sin nx)/(1 + nx))$ for $x \in \mathbb{R}$, $x \geq 0$.
6. Show that $\lim(\text{Arctan } nx) = (\pi/2)\text{sgn } x$ for $x \in \mathbb{R}$.
7. Evaluate $\lim(e^{-nx})$ for $x \in \mathbb{R}, x \geq 0$.
8. Show that $\lim(xe^{-nx}) = 0$ for $x \in \mathbb{R}$, $x \geq 0$.
9. Show that $\lim(x^2e^{-nx}) = 0$ and that $\lim(n^2x^2e^{-nx}) = 0$ for $x \in \mathbb{R}$, $x \geq 0$.
10. Show that $\lim((\cos \pi x)^{2n})$ exists for all $x \in \mathbb{R}$. What is its limit?
11. Show that if $a > 0$, then the convergence of the sequence in Exercise 1 is uniform on the interval $[0, a]$, but is not uniform on the interval $[0, \infty)$.
12. Show that if $a > 0$, then the convergence of the sequence in Exercise 2 is uniform on the interval $[a, \infty)$, but is not uniform on the interval $[0, \infty)$.
13. Show that if $a > 0$, then the convergence of the sequence in Exercise 3 is uniform on the interval $[a, \infty)$, but is not uniform on the interval $[0, \infty)$.